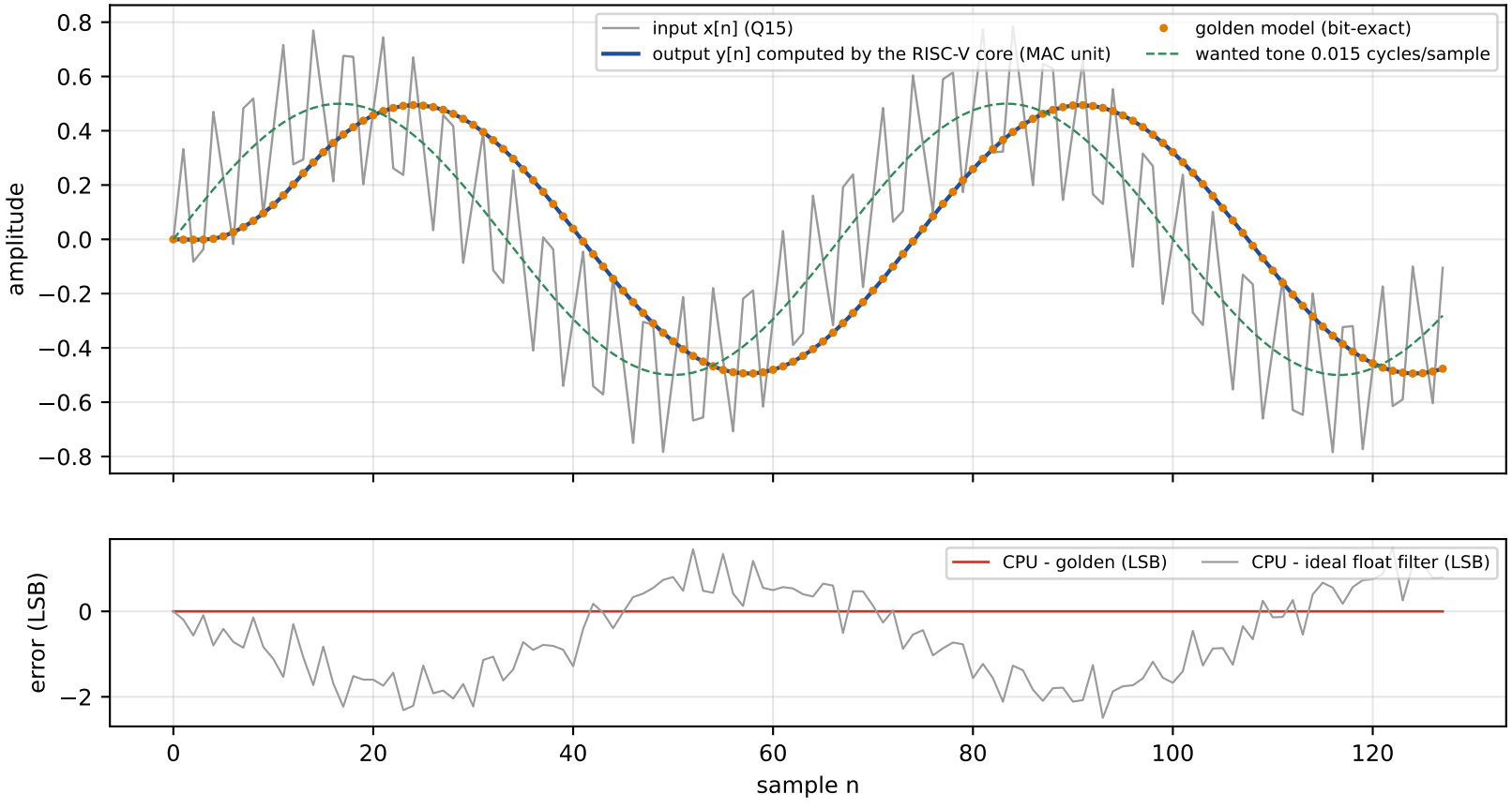
16-tap Q15 low-pass FIR executed on the processor (N = 128 samples)
input x[n] (Q15)
output y[n] computed by the RISC-V core (MAC unit)
golden model (bit-exact)
wanted tone 0.015 cycles/sample
amplitude
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
CPU - golden (LSB)
CPU - ideal float filter (LSB)
error (LSB)
0
−2
0
20
40
60
80
100
120
sample n